RAYHAN MUHAMMAD ALFARIZI

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Electrical Engineering graduate with an Information Technology concentration and hands-on experience in software engineering, IoT system, and machine learning. Experienced in developing web applications, APIs, database systems, and IoT solutions using Laravel, CodeIgniter, Python, and modern web technologies. Interested in developing reliable, maintainable, and scalable backend systems to solve real-world problems.

EDUCATION

Diponegoro University | Bachelor of Electrical Engineering **2022 – 2026**

- Information Technology Concentration | GPA: 3.87 / 4.00
- Relevant Coursework: Algorithms and Programming, Data Structures, Object-Oriented Programming, Software Engineering, Database, Operating Systems, Web Development, Modern Programming Methods

EXPERIENCE

Software Developer Intern **Sep 2026 – Present**
PT Tiga Serangkai Inti Corpora

- Collaborated in a three-person team to maintain and continue a **SaaS** web application, responsible for managing and improving the CRM module developed by previous developers.
- Worked with a **Laravel** backend and **React** frontend in a monolithic architecture.

Teaching Assistant – Algorithms and Programming **Feb – Jun 2025**
Diponegoro University

- Assisted 30 students in **Python** and **C++** programming laboratory sessions.
- Guided programming fundamentals and debugging.

Web Developer Intern **Jan – Feb 2025**
PT Des Teknologi Informasi (DESNET)

- Designed and developed a Guest Management System from scratch with one team member using **CodeIgniter 4**, including the backend, frontend, and database at dashboard, visit history, and scheduling.
- Implemented authentication, role-based authorization, **JavaScript** for scheduling interactions, and a **RESTful API** for guest form submission.
- Collaborated using **Git/GitHub** through task allocation, code reviews, and feature integration.

PROJECTS

Autoencoder-Based Fault Detection & Diagnosis for Smart Precision Irrigation (Final Project) **Jan – Jul 2026**

- Designed and developed the system as part of a three-person team, contributing to machine learning, data integration, and monitoring dashboard.
- Used **ESP32** and **MQTT** for sensor data communication, **FastAPI** for backend processing, and **Keras** for Autoencoder-based anomaly detection, achieving **97.56% F1-score**.
- Used **InfluxDB** for time-series data, **MariaDB** for application data, and **Laravel** with **Livewire** for the monitoring dashboard; containerized and deployed services using **Docker** on a VPS.

Artikula: Reading Space (Independent) **Feb 2026**

- Developed a full-stack article publishing platform using **Laravel** and **Livewire** for application logic and interactive interfaces, with **Supabase PostgreSQL** for relational data management.
- Implemented authentication, article management, search, filtering, analytics, and feedback features.

Bubble Pop Game (Assignment) **Nov – Dec 2025**

- Built an interactive browser game using **React** and **TypeScript** for UI and game logic, and **Tailwind CSS** for responsive styling.
- Implemented real-time scoring, animations, pause functionality, and customizable difficulty settings.

Letsgobois Shop – E-Commerce Platform (Independent) **Sep – Dec 2025**

- Developed an e-commerce platform using **Next.js**, with **Firebase** and **Google Authentication** for user management and **Midtrans** for online payment processing.

SI PANDA – Sistem Pantau Data Anak (KKN) **Jul – Aug 2025**

- Developed a child growth monitoring application using **CodeIgniter 4** for application logic, **Supabase PostgreSQL** for database, and **Tailwind CSS** for styling.
- Implemented child data management, measurement records, search, data visualization, and monthly recaps.

SI JEMPANG – Sistem Jentik Pantau Kampung (KKN) **Jul – Aug 2025**

- Developed a mosquito larvae monitoring application using **CodeIgniter 4** and **Supabase** for data management, with **Leaflet.js** for location-based visualization and monitoring.

PUBLICATIONS & COPYRIGHT

- Autoencoder-Based Fault Detection and Diagnosis for IoT-Enabled Smart Precision Irrigation of Avocado Trees | Presented at IEEE ICITACEE 2026; awaiting publication
- Copyright: Program Komputer – Sistem Fault Detection and Diagnosis berbasis Autoencoder pada Smart Irrigation Precision | DJKI, 2026

LEADERSHIP

Head of the Committee – Elektro Forge (E-Forge) **Aug – Sep 2024**
Electrical Engineering Student Association (HME), Diponegoro University

- Led a 5-member organizing committee delivering a PCB Design Training for 54 participants.
- Coordinated planning, logistics, and stakeholder communication to ensure successful event execution.

TRAINING & CERTIFICATIONS

- Python Essentials 2 – Python Essentials 1 – C Essentials 1 | Cisco Networking Academy
- Machine Learning for Beginners — Data Analysis with Python — Basic SQL | Dicoding Indonesia

LANGUAGES

- Indonesian – Native
- English – Professional Working Proficiency
  SEU TOEFL Prediction Test (Equivalent Score: 513)